

Written in the stars: did your specialty choose you?

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ABSTRACT

Purpose of the study Established over 2000 years ago, horoscopes remain a regular feature in contemporary society. We aimed to assess whether there could be a link between zodiac sign and medical occupation, asking the question—did your specialty choose you?

Study design A questionnaire-based study was distributed using an online survey tool. Questions explored the zodiac sign, specialty preferences and personality features of physicians.

Results 1923 physicians responded between February and March 2020. Variations in personality types between different medical specialties were observed, introverts being highly represented in oncology (71.4%) and rheumatology (65.4%), and extroverts in sexual health (55%), gastroenterology (44.4%) and obstetrics and gynaecology (44.2%) ($p < 0.01$). Proportions of zodiac signs in each specialty also varied; for example, cardiologists were more likely to be Leo compared with Aries (14.4% vs 3.9%, $p = 0.047$), medical physicians more likely Capricorn than Aquarius (10.4% vs 6.7%, $p = 0.02$) and obstetricians and gynaecologists more likely Pisces than Sagittarius (17.5% vs 0%, $p = 0.036$). Intensive care was the most commonly reported second choice career, but this also varied between zodiac signs and specialties. Fountain pen use was associated with extroversion ($p = 0.049$) and gastroenterology ($p < 0.01$).

Conclusions Personality types vary in different specialties. There may be links to zodiac signs which warrant further investigation.

INTRODUCTION

Astrology is defined as the study of the movements and positions of the stars and planets, in relation to human affairs and events.¹ Subjects are grouped into different zodiac signs depending on the exact time of their birth, and the relationship between the earth, the sun and each constellation at that time.^{2,3} Early astrologists described personality traits and temperaments of people born within each of the 12 constellations (figure 1).⁴ In current times, these descriptions, as well as predictions of the life events the subjects may encounter, are referred to as horoscopes. Thought to have been established 2400 years ago in Babylon, horoscopes in some form have been a constant feature across many cultures and religions since.² Despite this, there have been no scientific studies supporting any of the astrological theories and thus it is considered a pseudoscience.⁵

In 1985, Carlson conducted a double-blind trial to assess whether astrologers could match subjects to their zodiac sign.⁶ The results showed no difference compared with chance, leading the authors to conclude that there was no association between

an astrologers' predicted connection between the positions of the planets and other astronomical objects at the time of birth, and the personalities of test subjects. Astrologists proposed a number of potential reasons for the lack of positive findings; however, the scientific community accepted this experiment as evidence and refuted the astrological hypothesis.⁶ Despite this work, the popularity of the horoscope remains. In the current day, the majority of popular newspapers contain horoscope-based predictions for their readers,⁷ and astrology books regularly feature in the bestselling books list.⁸ A national foundation survey from 1999 found 12% of Americans read their horoscope every day,⁷ and a more recent poll reported 25% of Americans and Canadians and 24% of Britons believe in astrology.⁹

Within clinical medicine, different zodiac signs have been linked to different specialties. High pressured procedural-based specialties including surgery and cardiology have been linked to 'fire' signs such as Leo, while careers focused on discussion and contemplation, such as psychiatry and neurology, have been linked to 'air' signs such as Gemini and Aquarius.¹⁰ However, within each specialty, wide variations in job role may exist and therefore such generalisations are often dismissed.¹¹ Sievert *et al* examined personality-related factors that motivated individuals to seek a career in medicine and choose a specialty.¹² They found that an 'investigative' personality would likely choose internal medicine or pathology, those with a 'commanding' personality appeared to prefer surgery, while the 'rescuers' chose emergency medicine. The 'dependable' sought paediatrics while the 'compassionate' chose psychiatry.¹²

More classically, selecting a career is thought to be dependent on a number of factors, including previous clinical exposure, role models, work life balance¹³ and perceived suitability of one's personality characteristics to a certain career.¹⁴ If the described personality traits set out for each zodiac sign are correct (figure 1), associations between birth date and career choice should not be surprising.

To date, there is a paucity of data evaluating associations between zodiac signs and specialty choice. This study aimed to address this, asking: did you choose your specialty or did your specialty choose you?

METHODS

This population-based, cross-sectional study used an electronic survey to explore the horoscope, specialty preferences and personality features of physicians. The survey was developed by three medical professionals from different specialties, piloted with 10 participants, adapted and finalised (online supplemental appendix 1). Adaptive questioning and



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Figure 1 Described personality traits of each star sign.³

branching logic were used; the total number of questions therefore varied from 10 to 14 depending on participants' earlier responses. Questions covered the broad domains of zodiac sign, self-reported personality type, pen usage, vehicle choice, current medical specialty and a second-choice medical specialty. Participants were restricted to a single answer per question.

The online platform Google Forms hosted the survey, which was open with no password requirement. The survey was distributed via email to UK hospital postgraduate centres and posted on social media. Completion of the survey was voluntary without any form of remuneration for participating. The survey was open for a total of 6 weeks. All qualified physicians in the UK were eligible for inclusion. Any other allied healthcare professional and students were excluded. The study was discussed with the local Research Ethics Committee and exempt from formal approval as no individually identifiable data were collected.

Data were collated in Google Sheets software and transferred to Statistical Package for the Social Sciences (SPSS) V.26.0 (IBM Corp.) for statistical analysis. Postgraduate year 1 and 2 doctors (foundation) were excluded from specialty-specific analysis as it was felt they are unlikely to have made permanent career decisions at this stage. Data are presented as absolute numbers and percentages. Independent samples t-test, χ^2 and Fisher's exact tests were performed where appropriate to evaluate associations between variables.

RESULTS

Demographics

Nineteen hundred fifty-eight responses were recorded over the 6-week study period. Response rate could not be calculated as the total number of distributees was unavailable. Thirty-five

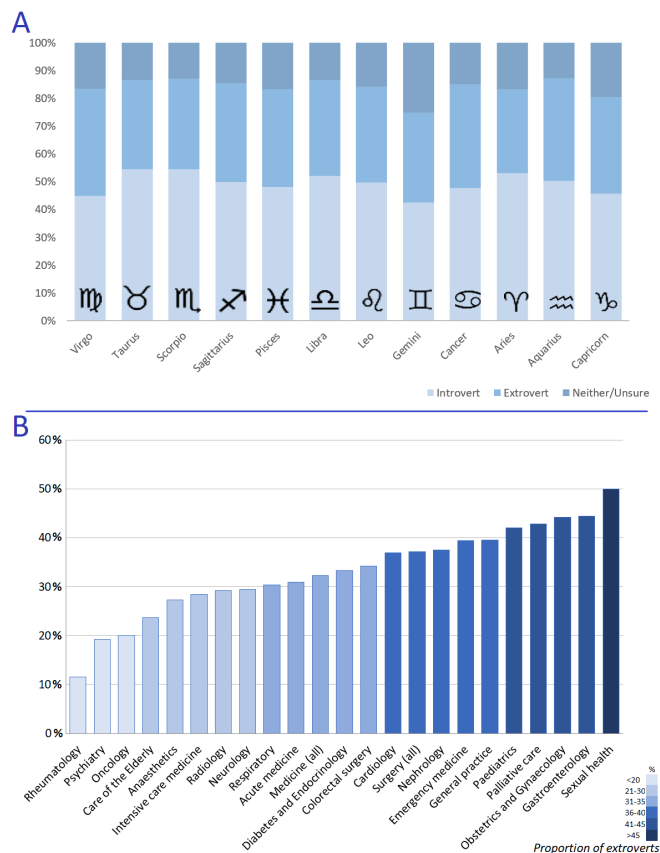


Figure 2 (A) Proportions of introverts and extroverts within each star sign group. (B) Percentage of respondents who identified as an extrovert in each specialty group.

responses were removed as they were identified as inappropriate or incomplete; 1923 responses were included in the analysis.

There was a slight female predominance with a male-to-female ratio of 1:1.33 (817 male, 1098 female, 8 other). The mean age respondents became a consultant/general practitioner (GP) were 35.2 years across all zodiac signs (range of means 34.7 years (Pisces)–36.7 years (Libra)); there was no statistically significant difference in age at achieving consultancy between the zodiac signs ($p=0.055$) (online supplemental table 1). Senior doctors contributed the majority of responses, with 587/1923 (30.5%) hospital consultants, 72/1923 (3.7%) general practitioners (GPs), 33/1923 (1.7%) staff grade/associate specialist and 675/1923 (35.1%) specialist registrars; there was an equal variation of grades across each zodiac sign. Between 138 and 187 responses were received from each zodiac sign. The highest number of responses was received from Virgos (187/1923, 9.7%) while the lowest from Capricorns (138/1923, 7.8%).

For all consultants and specialty doctors, 44.5% reported working in a district general hospital (range 33.3 Virgo–58.5 Aquarius) and 43.2% reported working in a tertiary hospital (31.7 Taurus–60.3 Virgo) (online supplemental table 1). The remaining listed either community or in an academic setting.

Personality features

49.5% of respondents considered themselves introvert, 34.4% extrovert and 16% neither/prefer not to say. There was no significant variation in introverts versus extroverts across zodiac signs ($p=0.412$) (figure 2A). The zodiac signs with the highest proportion of introverts were Scorpio and Taurus (54.5%) while

Table 1 Survey responses shown according to zodiac sign and medical specialty

	Virgo	Taurus	Scorpio	Sagittarius	Pisces	Libra	Leo	Gemini	Cancer	Aries	Aquarius	Capricorn	Total	P value (χ^2)
Total number	187	165	178	138	156	157	153	176	180	146	149	138	1923	
Scale-up value	1	1.13	1.05	1.36	1.2	1.19	1.22	1.06	1.04	1.28	1.26	1.36		
Medicine	67	62.3	64.1	56.9	61.1	59.6	59.9	68	59.2	56.4	50.2	78.6	754.3	0.021*
Surgery	15	14.7	13.7	14.9	15.6	11.9	17.1	20.2	20.8	17.9	22.6	16.3	211.6	0.855
Emergency medicine	6	15.9	14.7	13.6	6	9.5	12.2	14.9	13.5	14.1	8.8	16.3	156.4	0.471
Intensive care medicine	8	14.7	7.4	12.2	6	7.1	13.4	7.4	5.2	10.2	15.1	8.1	125.9	0.326
Anaesthetics	13	17	22.1	23	18	27.4	18.3	14.9	16.6	21.8	15.1	16.3	234.4	0.133
General practice	12	7.9	10.5	16.3	19.2	10.7	6.1	12.8	15.6	11.5	16.3	9.5	159.4	0.054
Cardiology	16	10.2	12.6	13.6	18	21.4	28.1	12.8	16.6	7.7	11.3	16.3	195.5	0.047*
Obstetrics and gynaecology	3	4.5	5.3	0	10.8	2.4	1.2	5.3	4.2	3.8	7.5	2.7	61.7	0.036*
Radiology	2	2.3	3.2	4.1	1.2	0	4.9	2.1	3.1	2.6	1.3	1.4	39	0.815
Paediatrics	14	13.6	8.4	9.5	13.2	11.9	6.1	10.6	12.5	7.7	7.5	12.2	138.2	0.822
Respiratory medicine	5	10.2	7.4	6.8	7.2	2.4	4.9	5.3	6.2	7.7	3.8	5.4	83.2	0.818
Gastroenterology	8	4.5	8.4	5.4	4.8	1.2	0	7.4	5.2	2.6	5	2.7	66.3	0.210
Acute medicine	5	4.5	4.2	4.1	6	1.2	0	4.3	5.2	5.1	1.3	5.4	57.2	0.631
Trauma and orthopaedics	3	2.3	0	2.7	4.8	2.4	2.4	3.2	2.1	3.8	7.5	2.7	47.9	0.506
Oncology	4	1.1	4.2	2.7	2.4	3.6	1.2	2.1	5.2	5.1	3.8	2.7	49.2	0.912
Otolaryngology (ENT)	2	1.1	1.1	0	3.6	1.2	0	5.3	2.1	2.6	1.3	4.1	35.2	0.212
Care of the elderly	9	10.2	6.3	4.1	2.4	8.3	3.7	14.9	2.1	10.2	3.8	6.8	92.7	0.024*
Psychiatry	3	4.5	5.3	4.1	7.2	7.1	2.4	5.3	6.2	7.7	7.5	1.4	72.8	0.713
Palliative care	0	3.4	3.2	0	3.6	0	2.4	4.3	1	1.3	2.5	1.4	34	0.397
Rheumatology	3	3.4	0	6.8	0	0	1.2	2.1	6.2	3.8	2.5	1.4	41.5	0.049*
Sexual health	1	1.1	0	4.1	1.2	2.4	3.7	2.1	2.1	6.4	2.5	2.7	40.3	0.411
Colorectal surgery	5	5.7	4.2	2.7	0	6	4.9	0	6.2	1.3	2.5	1.4	50.8	0.171

While statistical analysis was conducted on raw data, responses shown here have been scaled to even response trends across star signs, as shown in row 3. For each specialty, the red box highlights the most prevalent star sign, and blue shows the least prevalent.

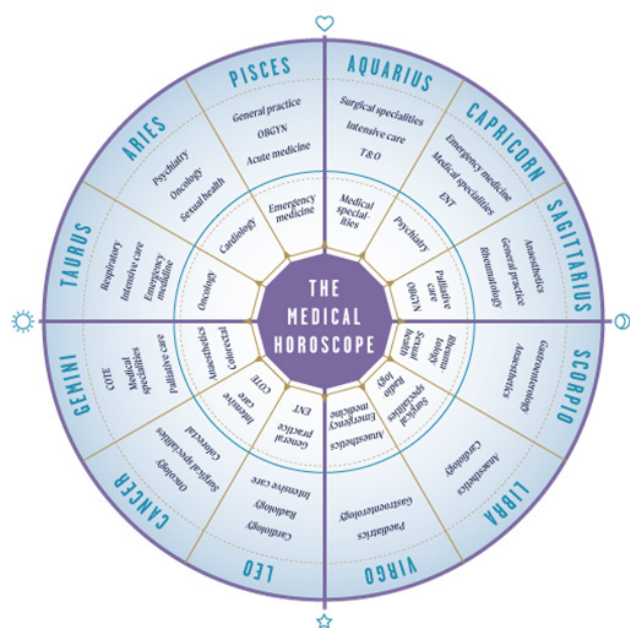


Figure 3 The most prevalent (outer circle) and least prevalent (inner circle) medical specialties in each zodiac sign. COTE, care of the elderly; ENT, otolaryngology; OBGYN, obstetrics and gynaecology; T&O, trauma and orthopaedics.

the zodiac sign with highest proportion of extroverts was Virgo (38.5%). In relation to medical specialty, the specialty with the highest proportion of introverts was oncology (71.4%) followed by rheumatology (65.4%), and the specialty with highest number of extroverts was sexual health (55%) ($p < 0.01$) (figure 2B; online supplemental table 2).

Across all respondents 88.87% felt their personality was suited to their chosen specialty: the highest being neurologists, and those in sexual health (100%), with the least suited being surgeons (77.1%) (online supplemental table 2). There was a statistically significant difference between the zodiac sign group who felt their personality suited their specialty the most (Taurus, 90.4% of respondents) and the zodiac sign group who felt their personality was least suited to their specialty (Leo, 82.5% of respondents) ($p = 0.036$) (online supplemental table 1).

Medical specialties

A number of trends were evident within responses, shown in table 1. Notably, a higher proportion of cardiologists were Leo compared with Aries (14.4% vs 3.9%, $p = 0.047$), more medical physicians were Capricorn compared with Aquarius (10.4% vs 6.7%, $p = 0.02$), and obstetricians and gynaecologists were more likely to be Pisces than Sagittarius (17.5% vs 0%, $p = 0.036$). Furthermore, rheumatologists were more likely to be Sagittarius than Scorpio or Pisces (16.4% vs 0%, $p = 0.049$), and care of the elderly physicians was more likely Gemini than Cancer (16.1% vs 2.3%, $p = 0.024$). Trends were also seen in GPs (12.0% Pisces, 3.8% Leo), anaesthetists (11.7% Libra, 5.5% Virgo) and colorectal surgeons (Cancer 12.2%, Gemini and Pisces 0%) but these did not reach statistical significance. Figure 3 displays the most prevalent (outer circle) and least prevalent (inner circle) medical specialties in each zodiac sign.

Alternative specialty choice

The most common second choice specialty was intensive care medicine (103/1329, 7.8%). Other popular choices included anaesthetics (6.2%), emergency medicine (5.5%) and general practice (5.0%).

Of 1329, 69 (5.1%) respondents reported that if they could not practice their current specialty, they would not want to do anything else. Notable answers for a second career choice included disc jockey, artist, McDonald's worker, househusband, horticulturist, marine biologist, cocktail drinker, cabinet maker, internet millionaire, yoga teacher, mountain climber and bookshop owner. Second choice by zodiac sign is shown in online supplemental table 2. Most popular second choice specialty for each specialty group is depicted in figure 4.

Pen usage

Participants were asked what type of pen they used to write with at work. Of the respondents who reported using a fountain pen, 52.6% identified as being an extrovert and 47.4% identified as being introvert. There was a significant association between identifying as an extrovert and using a fountain pen versus a biro/roll-erball ($p = 0.049$). When reviewed by specialty, gastroenterology had the highest proportion of fountain pen users (21.2%, when compared with the remaining respondents $p < 0.01$), followed by acute medicine (11.9%) (figure 5). The zodiac sign with the highest fountain pen usage was Taurus (19.1%, compared with all respondents $p = 0.14$) and the lowest Leo (3.2%).

DISCUSSION

This is the first study looking at zodiac signs and medical career choices. Despite being considered a pseudoscience,⁵ zodiac signs and their links to personality characteristics remain popular topics of discussion^{7,9} even among the scientifically minded medical community. In a 6-week period our survey drew 1958 responses, and generated high levels of interest on the social media platform Twitter.

No significant differences were seen between proportions of introverts and extroverts in each zodiac sign ($p = 0.412$); however, variations were seen between medical specialties (figure 2). Studies have interrogated links between personality-type and medical careers.^{14,15} Surgeons tend to exhibit high levels of 'impulsive sensation seeking' behaviour,¹⁶ and alongside other procedural-based medical specialties, such as cardiology, are more commonly extroverted and 'thinking types' on Myer Briggs analyses.¹⁷ 'Introversion and feeling' types are more likely to choose primary care.¹⁷ Contrary to Stilwell, our study found non-surgical specialties had the highest proportion of self-reported extroverts: sexual health, gastroenterology and obstetrics and gynaecology (44.2%–50.0% extrovert) (figure 5). We found that less acute, more outpatient-based specialties including rheumatology, psychiatry and oncology had the highest levels of introversion. Anaesthetists were predominantly introvert (59.8%), in keeping with previous work.¹⁸ This paper would support the use of personality-based specialty advice tools in aiding junior doctors in making career choices.¹⁹

Across zodiac signs, significant differences were found between how well physicians felt their personality suited their career, with 90.4% of Taurus agreeing but only 82.5% of Leos ($p = 0.036$). It could be argued that this finding reflects the typical personality traits associated with the respective star signs; the persistent determined nature of Taurus may demand well-matched specialty-specific qualities, while the intolerant Leo may experience discontent with their specialty choice (figure 1).



There have been several robust studies looking at month of birth and educational attainment.²⁰ A well-documented spike exists in educational success between children born in autumn months when compared with their summer counterparts. Given those born in autumn are almost a full year older than those born

Intensive care was the most popular overall second choice specialty (7.8%). This is in contrast to other work, which suggests one of the most important factors in doctors choosing a specialty

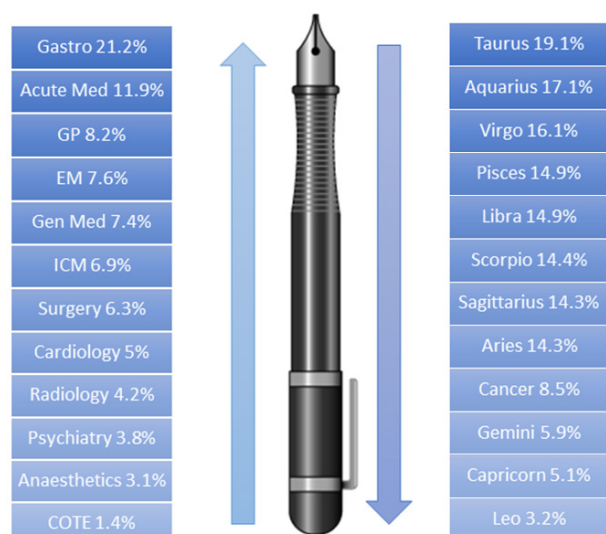


Figure 5 Fountain pen use across specialties and zodiac signs. COTE, care of the elderly; EM, emergency medicine; GP, general practice; ICM, intensive care medicine.

is hours worked on call.¹³ With increasing moves towards senior presence and 24/7 working on both intensive care units and in other acute areas of medicine, these specialties have been found to work longer and more antisocial hours.²⁴ However, we speculate that the recent COVID-19 pandemic may have had an impact on physicians' second choice preference, recent new coverage surrounding intensive care may have highlighted the specialty and increased its prestige in the medical community.

Group trends were consistently shown for second choice career options in each specialty (figure 4). These often involved specialties with overlap, including intensive care medicine with anaesthetics, haematology with pathology and oncology with palliative care. Interestingly 5.1% stated they would not wish to do any other job than their current specialty; many felt if they could not do their current specialty, they would instead leave medicine altogether.

A significant association between extroversion and fountain pen use was identified ($p=0.049$). Research has shown that extroverts are more likely to have a desire for social ambition and place value on status.²⁵ We contend that compared with its counterpart, the biro, a fountain pen may be considered a high status, luxury item within the realm of medicine, corroborating existing evidence that extroverts are more likely to splurge.²⁶ With regards to specialty, gastroenterologists were avid fountain pen users compared with all other specialties (21.2%, $p<0.01$) (figure 5); almost double the 11.5% of internal medicine subspecialties found by Zimerman *et al* to drive a luxury car²⁷; cardiologists were the most likely specialty to drive a luxury car in that work²⁷; however, only 5% used a fountain pen in our cohort. Perhaps cardiologists focus more on their appearance driving into the carpark than on the ward?

Several limitations are to be considered in this study; a voluntary social media distributed questionnaire introduces a selection bias, compounded moreover by the timing during the COVID-19 pandemic; possibly affecting those specialties more available to participate and potentially influencing the second choice of career. Due to the distribution and categorical nature of data, including varying numbers of responses for each specialty, statistical analysis was limited. We acknowledge this

What is already known

- Medical specialty choices are often associated with certain personality features.
- Personality traits and temperaments have also been associated with people born within each of the 12 zodiac signs. No prior study has examined zodiac signs and their correlation with medical specialties.

Main messages

- Associations appear to exist between zodiac signs and medical specialties.
- Varying personality types exists between different specialties, with the highest levels of extroversion found in sexual health, gastroenterology and obstetrics and gynaecology, and highest levels of introversion in rheumatology and oncology.
- Fountain pen use is highest in extroverts, as well as gastroenterologists.
- These results may aid in career planning.

may reduce the number of firm conclusions that can be drawn from the data. Self-reporting of personality traits may introduce response bias where specialties may adapt their reporting of characteristics in order to fit the pre-existing ideals of what 'should be'. Well recognised in the study of astrology, confirmation bias is common where results are interpreted and reported in such a way to confirm a preconceived outcome. Horoscopes remain popular when there is little scientific research to support the theories⁶; one explanation for this is the 'Barnum effect'.²⁸ When vague and generalisable information regarding personalities is presented, we ignore predictions that do not fit and focus on predictions which do, internally giving the information high accuracy ratings.²⁸ Some studies suggest that those who read horoscopes will look for 'illusory correlations' between predictions and real life events.²⁹ It must also be considered that some physicians may not be working in the specialty of their preference, as career choice is affected by availability of positions as well as desire to work in one field.

CONCLUSIONS

In a large-scale study assessing links between zodiac signs and medical specialties, links between specialties and personality types were identified which corroborate previous research. Trends clearly exist between zodiac signs and some specialties however larger numbers per zodiac and specialty group would be required to prove significance. Ultimately, it appears the zodiac sign's impact on career decision remains a 'pseudoscience', for now at least.

Twitter Holly Morgan @hollymorgs

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Contributors HM planned the study, developed and distributed the survey, analysed results and produced the manuscript. HC researched the background, distributed the survey, aided in manuscript and figure production. SM reviewed the data, carried out statistical analysis, edited the manuscript. CE developed and distributed the survey, aided in manuscript production and editing.

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